

Right Angles & Trigonometry

US Common Core – High School Geometry

Domain: Similarity, Right Triangles, and Trigonometry (G-SRT.B, G-SRT.C)

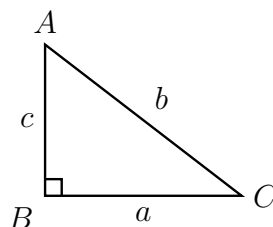
1 The Pythagorean Theorem

Theorem

If $\triangle ABC$ is a right triangle with the right angle at B , then

$$a^2 + c^2 = b^2$$

where b is the hypotenuse (opposite the right angle).



Converse of the Pythagorean Theorem

For any $\triangle ABC$, if $a^2 + c^2 = b^2$, then $\triangle ABC$ is a right triangle (right-angled at B).

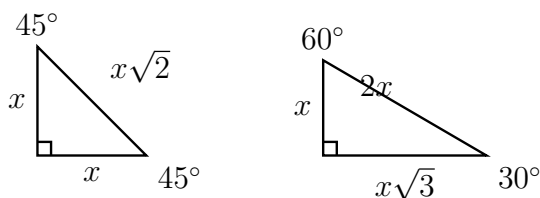
2 Special Right Triangles

45°-45°-90° Triangle

Legs of length x ; hypotenuse $x\sqrt{2}$ (from $x^2 + x^2 = 2x^2$).

30°-60°-90° Triangle

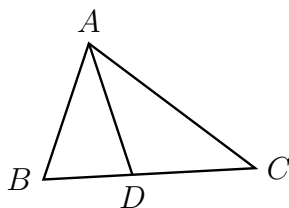
Short leg (opposite 30°) of length x ; long leg (opposite 60°) of length $x\sqrt{3}$; hypotenuse $2x$.



3 Altitude to the Hypotenuse

Theorem (Right Triangle Altitude Similarity)

If $\triangle ABC$ is a right triangle and an altitude is drawn from the right angle to the hypotenuse, then the two smaller triangles formed are similar to each other and to the original triangle.



Geometric Mean

The geometric mean of two positive numbers a and b is $x = \sqrt{ab}$.

Geometric Mean (Altitude) Theorem

The length of the altitude to the hypotenuse is the geometric mean of the two segments it creates on the hypotenuse:

$$BD^2 = CD \cdot AD$$

Geometric Mean (Leg) Theorem

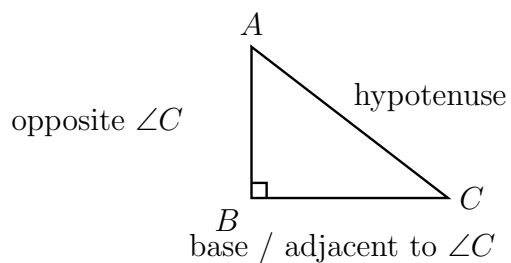
The length of a leg of the right triangle is the geometric mean of the whole hypotenuse and the segment of the hypotenuse adjacent to that leg:

$$CA^2 = DC \cdot BC, \quad AB^2 = BD \cdot BC$$

4 Trigonometric Ratios

Naming Sides Relative to an Angle

For an acute angle (e.g. $\angle C$) in a right triangle: the side opposite the right angle is the **hypotenuse**; the side opposite $\angle C$ is the **opposite** side; the remaining side is the **adjacent** (or base) side.



SOH-CAH-TOA

$$\sin C = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{AB}{AC} \quad \cos C = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{BC}{AC} \quad \tan C = \frac{\text{opposite}}{\text{adjacent}} = \frac{AB}{BC}$$

Note: Also, $\tan \theta = \frac{\sin \theta}{\cos \theta}$ for any angle θ .

Example

Right triangle with legs $AB = 7$, $BC = 8$, hypotenuse $AC = 9$ (right angle at B):

$$\tan C = \frac{7}{8}, \quad \sin C = \frac{7}{9}, \quad \cos C = \frac{8}{9}$$

5 The Law of Sines

Law of Sines

For any triangle with angles A, B, C and opposite sides a, b, c :

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Example

Triangle with $\angle A = 32^\circ$, side $b = 8$ (opposite B), side $a = 7$ (opposite A). Find $\angle B$:

$$\frac{\sin 32^\circ}{7} = \frac{\sin B}{8} \Rightarrow \sin B = \frac{8 \sin 32^\circ}{7} \approx \frac{8(0.53)}{7} = \frac{106}{175} \approx 0.605$$

$$B = \sin^{-1}(0.605) \approx 37.3^\circ$$

6 The Law of Cosines

Law of Cosines

For any triangle with sides a, b, c and angle C opposite side c :

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} \iff a^2 + b^2 - 2ab \cos C = c^2$$

Note: Use the Law of Cosines (not the Law of Sines) when all three sides are known (SSS) but no angle is known – the Law of Sines needs at least one known angle-side pair to get started.

Example (SSS triangle)

Triangle with $a = 7$, $b = 8$, $c = 9$. Find $\angle C$:

$$\cos C = \frac{7^2 + 8^2 - 9^2}{2(7)(8)} = \frac{49 + 64 - 81}{112} = \frac{32}{112} = \frac{2}{7} \approx 0.286$$

$$C = \cos^{-1}(0.286) \approx 73.4^\circ$$

Note: The source notes compute this final step as $\cos^{-1}(28/121) \approx 76.62^\circ$. The correct numerator and denominator from the Law of Cosines formula are 32 and 112 (not 28 and 121) – likely a slip when copying the fraction. The value shown above, $\approx 73.4^\circ$, uses the corrected arithmetic.